

19. Analytical/experimental solutions for internal flow HT

[Thermally fully developed flow]

$$h \neq f(x) \rightarrow Nu \neq f(x)$$

<Laminar flow>

- Analytical solutions can be obtained by solving the governing equations: $(x, r) \sim (u, v)$

$$\frac{\partial u}{\partial x} + \frac{1}{r} \frac{\partial(rv)}{\partial r} = 0$$

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial r} \right) = -\frac{dp}{dx} + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial r} = \frac{\alpha}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right)$$

- BC:

$$\text{At } r = r_0, u = v = 0 \text{ and } q_s'' = \text{const or } T_s = \text{const}$$

$$\text{At } r = 0, \frac{\partial u}{\partial r} = 0 \text{ and } \frac{\partial T}{\partial r} = 0 \text{ (} T \text{ finite)}$$

- In fluid mechanics, FD (hydrodynamic) $\rightarrow \frac{\partial u}{\partial x} = 0$ and Continuity EQ. $\rightarrow v = 0$

$$\text{Momentum EQ.} \rightarrow \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = \frac{dp}{dx}$$

$$\text{Sec. 8.1.3} \rightarrow u(r) = -\frac{1}{4\mu} \left(\frac{dp}{dx} \right) r_0^2 \left[1 - \left(\frac{r}{r_0} \right)^2 \right] \text{ or } u(r) = 2u_m \left[1 - \left(\frac{r}{r_0} \right)^2 \right]$$

- Substitution of $u(r)$ and $v=0$ into the energy equation and using the thermally FD flow

$$\text{condition for } q_s'' = \text{const}, \frac{\partial T}{\partial x} = \frac{dT_m}{dx} \text{ yields}$$

$$2u_m \left[1 - \left(\frac{r}{r_0} \right)^2 \right] \frac{dT_m}{dx} = \frac{\alpha}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \text{ for } q_s'' = \text{const}$$

Solving the above equation with BCs (T finite at $r=0$ and $T=T_s$ at $r=r_0$) gives

$$T(r, x) = T_s - \frac{2u_m r_0^2}{\alpha} \left(\frac{dT_m}{dx} \right) \left[\frac{3}{16} - \frac{1}{4} \left(\frac{r}{r_0} \right)^2 + \frac{1}{16} \left(\frac{r}{r_0} \right)^4 \right]$$

$$T_m = T_s - \frac{11}{48} \frac{u_m r_0^2}{\alpha} \left(\frac{dT_m}{dx} \right)$$

Substituting $\frac{dT_m}{dx} = \frac{q_s'' \pi D}{\dot{m} c_p} = \frac{q_s'' \pi D}{\rho u_m (\pi D^2/4) c_p} = \frac{4 q_s''}{\rho u_m D c_p}$ into the above equation yields

$$T_m - T_s = -\frac{11}{48} \frac{q_s'' D}{k}$$

Finally,

$$h = \frac{q_s''}{T_s - T_m} = \frac{48}{11} \frac{k}{D} = 4.36 \frac{k}{D} \rightarrow \boxed{Nu_D = 4.36} \text{ for } q_s'' = \text{const}$$

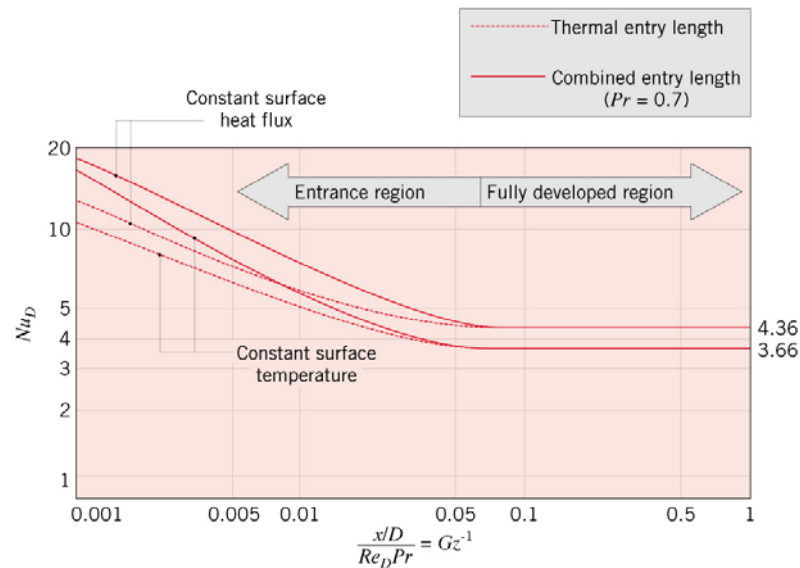
- Similarly, $\boxed{Nu_D = 3.66}$ can be obtained for $T_s = \text{const}$
- Details of derivation are in Sec. 8.4.1

(Note)

$Nu = \text{const} \rightarrow h = \text{const} \times k/D$ and $h \neq f(u_m)$ for FD flow. Why?

This explains why microchannel heat exchangers are more effective than conventional heat exchangers

In thermal entry region, $Nu = f(\text{Re}, \text{Pr})$



<Turbulent flow>

- Dittus-Boelter equation: $Nu_D = 0.023 Re_D^{4/5} Pr^n$

$0.7 \leq Pr \leq 160$
 $Re_D \geq 10,000$
 $L/D \geq 10$

$n=0.4$ for heating ($T_s > T_m$)
 $n=0.3$ for cooling ($T_s < T_m$)

Valid both for $q_s'' = \text{const}$ and $T_s = \text{const}$ but for small temp. difference

Properties evaluated at T_m

- For large property variation, Sieder and Tate equation

$$Nu_D = 0.027 Re_D^{4/5} Pr^{1/3} \left(\frac{\mu}{\mu_s} \right)^{0.14} \quad \begin{array}{l} 0.7 \leq Pr \leq 16,700 \\ Re_D \geq 10,000 \\ L/D \geq 10 \end{array}$$

All properties except μ evaluated at T_m

- There are many more correlations

[Thermal entry region]

For short tube lengths, correlations for thermal entry region required: $h = f(x) \rightarrow Nu = f(x)$

- For laminar flow, there are many correlations using Graetz number $Gz_D \equiv \frac{Re_D Pr}{(x/D)}$
- For turbulent flow, $\frac{Nu_D}{Nu_{D,fd}} = 1 + \frac{C}{(x/D)^m}$ (properties at $\bar{T}_m = \frac{T_{mi} + T_{mo}}{2}$)

Note $Nu_D \rightarrow Nu_{D,fd}$ as x becomes large

[Noncircular tubes]

- Use effective diameter and keep using the correlations for circular tubes
(Def.) Hydraulic diameter

$$D_h \equiv \frac{4A_c}{P} \quad (A_c: \text{tube cross sectional area, } P: \text{wetted perimeter})$$

- Reasonable to a first approximation but not accurate at corners
- Especially, less accurate for laminar flow and for tubes with sharp corners

Table 8.4 Summary of convection correlations for flow in a circular tube!